

Foundation (Jalapeno)

Q1. Simplify $\sqrt{16}$

- (A) 2
- (B) 4
- (C) 6
- (D) 8
- (E) 10

Q2. Simplify $\sqrt[3]{27}$

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q3. A stock's value increased by $\sqrt{225}$ dollars. What is the increase?

- (A) 10
- (B) 15
- (C) 20
- (D) 25
- (E) 30

Q4. Simplify $\sqrt{64} \cdot \sqrt{9}$

- (A) 6
- (B) 12
- (C) 18
- (D) 24
- (E) 30

Q5. A company's savings account earns $\sqrt[3]{125}$ dollars in interest monthly. How much does it earn?

- (A) 4
- (B) 5
- (C) 6
- (D) 7
- (E) 8

Q6. Simplify $\sqrt{121}$

- (A) 10
- (B) 11
- (C) 12
- (D) 13
- (E) 14

Q7. Express $\sqrt{x}$ as a rational exponent

- (A) $x^{1/2}$
- (B) $x^{1/3}$
- (C) $x^{2/3}$
- (D) $x^{3/4}$
- (E) $x^{4/5}$

Q8. Simplify $\sqrt{100} \cdot \sqrt{16}$

- (A) 20
- (B) 30
- (C) 40
- (D) 50
- (E) 60

Open Ended Questions (FRQ)

Q9. A tax consultant needs to calculate the square root of a client's total tax liability, $\sqrt{400}$. Simplify and calculate the value. Show all steps.

Q10. A business owner invests $\sqrt[3]{1000}$ dollars in a new venture. Express this value as a rational exponent and simplify. Show all steps.

Chef's Tips

- Tip 1: Ensure students simplify radicals correctly
- Tip 2: Remind students to convert between radical and rational exponent forms accurately
- Tip 3: Encourage students to show all steps for free response questions